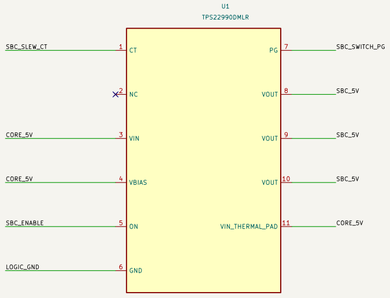
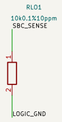
# HARDWARE-SUPERVISED SBC POWER — 3 A BRANCH CANDIDATE

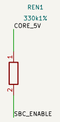
CORE_5V charges first. Supervisor release and controlled switch slew start SBC_5V without host software.
U1 pad11 is VIN, never GND. U2 has six terminals and no exposed thermal pad. All footprints remain unassigned.

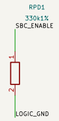


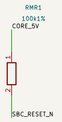






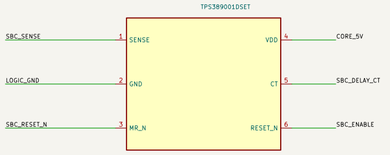














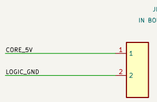


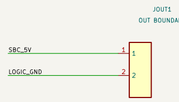


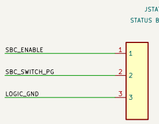










JMR1 accepts an optional normally-open dry contact to ground. No external power or active-high driver.
JSTATUS1 is a 5 V-domain diagnostic boundary, not direct 3.3 V GPIO. Switch PG means enhanced FET, not valid output voltage.







No branch current limiting, reverse blocking or arbitrary-brownout hold-up is supplied by this block.
CTSW1 requires >=470 nF effective; CTSUP1 >=1.2 uF effective. Values and timing require qualified parts/tests.
SBC total charge capacitance <=1 mF: host current <=3 A. No alternative USB/SBC power source until backfeed is reviewed.
No machine permission follows from power return. Circuit ERC is not PCB or physical qualification.

| | |
|---|---|
| Sheet: / | |
| File: sbc_power.kicad_sch | |
| **Title: Sequenced SBC supply** | |
| Size: A2 | Date: 2026-09-28 |
| KiCad E.D.A. 10.0.4 | **Rev: r0.1** |
| | Id: 1/1 |